

Multi Log Comparison Using Directly-Follows Graphs

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Abstract. Process comparison provides vital insights for practitioners. Currently, some tools enable a visual comparison of two event logs for specific metrics. When more than two logs are compared, multiple comparisons increase the Type 1 Error rate; i.e., significant differences may be found by chance, rather than because they exist. To overcome this issue, the goal is to create a tool with an easy-to-use, understandable interface for end users who may not be familiar with process mining.

1 Introduction

This proposal outlines a software development project to visualize differences across multiple event logs in terms of control flow and execution times. In process mining [1], comparison is an important field. To illustrate this, consider the treatment process of patients recorded in an event log. The data can be split into multiple sublogs, for example, female and male patients. In the next step, the treatment processes can be compared, which may reveal that certain steps take longer or that some activities occur more frequently in a log. While this example compares only two groups, multiple groups may be present in practice, such as when analyzing the production processes of multiple plants. A ProM plug-in exists for comparing two event logs [2], but there is no similar approach for comparing more than two event logs. The project aims to develop a tool that enables the identification of statistically significant differences in the process by utilizing Directly-Follows Graphs (DFGs) as an abstraction.

2 Objective

The primary objective is to develop a user-friendly tool that allows for comparing multiple event logs. It is important that the tools can handle large amounts of data in a reasonable time. The tool should be in a dashboard format, and an overview is provided in Figure 1.

In the top left, variants are listed across all logs. The variants are sorted by overall occurrence. The user can select which variants should be considered for the analysis. The bar chart shows the occurrence distribution across all logs. In the example, three logs are present.

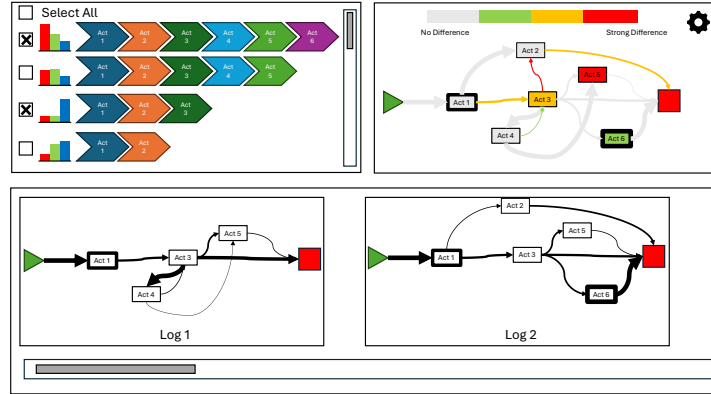


Fig. 1. Overview of the dashboard.

In the top right, a DFG is visualized based on the selected variants. The colors indicate whether a statistically significant difference exists between the logs with respect to the number of occurrences or the elapsed time between an event and its case’s start. The used metric can be changed in the settings. Thickness on arcs and nodes represents the frequency of its use when replaying the merged log. The elements should be clickable. When an element is selected, more information should be displayed, highlighting which event logs differ and how the distribution between the various logs looks.

At the bottom, DFGs for each individual log are displayed. It is important to note that the layout of each DFG should be the same as that of the merged DFG to facilitate easier comparison. The thickness of nodes and edges is determined for each log individually.

Crucial to the project is the use of statistical tests to check assumptions and detect differences. Only counting occurrences and highlighting differences will not help when considering different sample sizes. One first needs to run tests (e.g., the Shapiro-Wilk test) to assess the characteristics of the data. Then tests (e.g., a One-Way ANOVA or a Kruskal-Wallis test) are needed to determine whether differences exist. Afterwards, a pairwise post-hoc comparison (e.g., Tukey’s HSD, Dunn’s test with Bonferroni adjustment) is conducted to find out which logs/groups differ. To check how strong differences are and thus color the elements in the DFG, effect sizes (e.g., η^2 , ω^2) must be measured, and a coloring map must be created. When only two event logs are provided, the test setup must be adjusted.

References

1. van der Aalst, W.M.P.: Process Mining - Data Science in Action, Second Edition. Springer (2016). <https://doi.org/10.1007/978-3-662-49851-4>

2. Bolt, A., de Leoni, M., van der Aalst, W.M.P.: A visual approach to spot statistically-significant differences in event logs based on process metrics. In: CAiSE. pp. 151–166. Springer (2016). https://doi.org/10.1007/978-3-319-39696-5_10